Drone Delivery and Geographic Access to Medications

Research proposal for Planet Education and Research
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Research purpose

A pharmacy may be nearby on a map yet difficult to reach for someone without reliable transport. Prescription-drone services could change that constraint, but extending a delivery network does not establish that patients receive medication when they need it. I propose to study when eligibility for these services reduces the burden of obtaining medication and supports refill continuity.

The information systems question is how a new delivery capability becomes a usable resource within existing care. The proposed model links service eligibility, medication acquisition burden, and refill continuity, with baseline geographic disadvantage as a moderator. Planet imagery would characterize physical context alongside verified service histories and road-network measures.

Research questions

1. Access. Does operating-service eligibility reduce the burden of obtaining medication, particularly in geographically disadvantaged locations?

2. Mechanism. Is lower acquisition burden associated with improved refill continuity, and which barriers persist after service becomes available?

3. Measurement. Does historical imagery add reproducible information about physical context beyond documentary and road-network records?

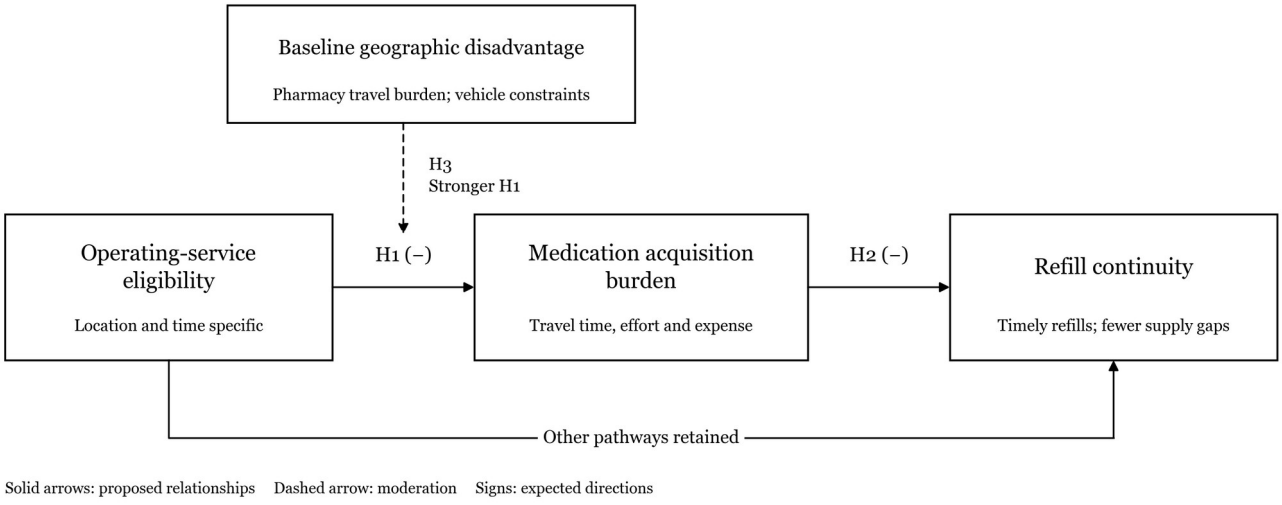


Figure 1. Proposed research model. H1 and H2 describe expected negative relationships; H3 predicts a stronger H1 under greater baseline geographic disadvantage. The residual pathway permits other mechanisms. These are hypotheses, not estimates; causal mediation requires additional data and identification assumptions.

Related literature and research gaps

Three research streams motivate the study of U.S. prescription delivery to patients at home.

What existing research establishes

Healthcare logistics. Haidari et al. [1] use simulation to show how drone vaccine transport can change supply availability and costs under different geographic and operational conditions. Nisingizwe et al. [2] document faster blood-product delivery and reduced wastage in Rwanda. These studies provide evidence about distribution systems; their settings and outcomes do not directly establish benefits for U.S. patients receiving prescriptions at home.

Medication access. Manne et al.'s recent scoping review [3] finds that the medication-delivery literature emphasizes feasibility, delivery performance, regulation, and implementation. Evidence on adherence, clinical outcomes, patient experience, and equity remains limited. Levesque et al. [4] conceptualize access as an interaction between service characteristics and people's abilities to obtain care, supporting a distinction between geographic availability and realized access.

Information systems mechanisms. Strong et al. [5] explain how the possibilities offered by an information system become organizational outcomes through affordance actualization. I use this as a theoretical lens: delivery capability creates a potential benefit that depends on pharmacy integration, eligibility, adoption, and local constraints. The original study concerns EHR implementation; applying the lens to drone delivery is part of this proposal.

Research gap and hypotheses

The unresolved issue is the conversion of delivery capability into usable access. Geographic disadvantage may increase the value of avoiding a pharmacy trip, while fees, digital access, prescription eligibility, and pharmacy coordination can still prevent use. The model therefore allows pathways beyond acquisition burden; it does not assume complete mediation.

Hypothesis	Proposed relationship
H1 Access mechanism	Operating-service eligibility reduces medication acquisition burden, relative to the options otherwise available.
H2 Continuity mechanism	Greater acquisition burden is associated with lower refill continuity. Reducing that burden is a proposed mechanism, not an established causal effect.
H3 Moderation	The negative effect in H1 is stronger where baseline geographic disadvantage is greater, measured before rollout.

Expected contribution

The study connects verified service exposure to a measurable access mechanism and the distribution of benefits. It asks when a digital delivery capability relaxes a binding constraint within existing care, and whether satellite imagery adds useful evidence to an auditable exposure registry.

Study setting and the role of Planet imagery

The initial search covers the contiguous United States during 2019–2026, beginning with College Station, Texas, and the Salt Lake Valley, Utah. I will seek approximately 12 months before and up to 24 months after verified starts, subject to archive availability and operating duration. Prescription delivery, general retail delivery, and medical-specimen transport will be coded separately.

A July 2024 report to the Texas pharmacy board records approval on 2 May 2023 and prescription-drone service beginning on 28 August 2023 [7]. Amazon's later public notice states that the College Station pilot ended, without specifying the closure date [8]. Intermountain announced service for selected eligible Salt Lake Valley households on 4 October 2022 [9]. These are documentary starting points; neither implies countywide or continuous coverage.

Distinct evidence for distinct constructs

Construct	Measurement and source
Service eligibility	Verified operating periods and household-level or geographically precise coverage from operator, pharmacy, or regulatory records.
Baseline disadvantage	Pre-rollout pharmacy travel burden and household vehicle availability; report these dimensions separately as well as any pre-specified index.
Acquisition burden	Observed travel time, effort, and expense from surveys or linked service records where available. Modeled road access is a proxy, not the mediator itself.
Refill continuity	Refill gaps or timely refills from longitudinal pharmacy or claims records. Filling a prescription does not demonstrate ingestion.

Requested imagery and pilot scope

I request historical PlanetScope orthorectified surface-reflectance imagery with metadata and quality masks. Its approximately 3 m output grid [10] supports a pilot assessment of facility-scale context; it should not be assumed to resolve individual drones, small pads, or delivery activity. If separately authorized, selected 50 cm SkySat imagery [11] could support validation. SkySat and tasking are not included in E&R Basic access [13].

The imagery pilot would cover up to 10 candidate and 10 comparison sites, each with a 4 km² area of interest and 12 dates: 960 km² of area-date requests. This is a sampling calculation, not a quota guarantee. Archive coverage, download accounting, and E&R terms [12] will be checked before ordering.

Registry and image validation

The registry will preserve source provenance and uncertainty in dates and coverage. Imagery checks will address cloud, shadow, registration, and seasonality. Two independent annotations and held-out locations will assess agreement and, where labels permit, precision and recall. Last-absent and first-present images bound physical change, not the start of an operating service.

Measurement and empirical strategy

Exposure and outcomes

The primary exposure is eligibility for an operating service at a location and time, based on verified coverage, pauses, and closure dates. Actual use is a subsequent behavioral choice; it will not replace eligibility when estimating availability effects. OSM road impedance, pharmacy access, Census demographics, and vehicle availability will be measured before rollout where possible.

The construct definitions in the preceding section determine the evidence required for each test. Modeled geographic access will be distinguished from observed acquisition burden and completed deliveries. Refill continuity requires longitudinal dispensing or claims records, which have not been secured. Clinical outcomes would be exploratory and require separate approved patient data; neither spending nor mobility changes establish a health benefit.

Identification and heterogeneity

If the registry yields enough independent rollout units and suitable longitudinal outcomes, I will estimate cohort-specific difference-in-differences and event studies following Callaway and Sant'Anna [6]. The estimand is the effect of service eligibility among eligible locations or patients. Never-treated or not-yet-treated units will be considered as comparisons, conditional on overlap and a defensible parallel-trends assumption.

I will examine pre-treatment trends, selective rollout, concurrent pharmacy or transport changes, seasonality, and spillovers. Inference will be clustered at the rollout decision level; a large number of small geographic cells does not create many independent interventions. Pre-specified heterogeneity will focus on baseline travel burden, pharmacy access, and vehicle availability. Exposure buffers and alternative coverage definitions will probe spillover and classification sensitivity.

Because service can pause or close, an absorbing-treatment estimator is appropriate only within verified uninterrupted operating windows. Exits and interruptions will be analyzed separately. Regulatory approval, weather, and distance will not be treated as automatically valid instruments; each could affect access or outcomes through other pathways.

Testing the mechanism

Eligibility effects on acquisition burden would address H1; interactions with pre-rollout disadvantage would address H3. The total eligibility effect on refill continuity is distinct from H2. A causal test of mediation would additionally require linked longitudinal mediator and outcome data, temporal ordering, and defensible assumptions about mediator–outcome confounding. Without these, I will report descriptive mechanism evidence and avoid a causal mediation claim. Post-treatment burden will not be included as a control in the total-effect model.

Limits on interpretation

Outcome access, historical coverage, and the number of independent deployments remain uncertain. With only a few usable sites, I will present a comparative feasibility study rather than a national causal estimate. Without linked outcomes, results will be restricted to service availability and modeled accessibility. No causal or clinical conclusion will be inferred from satellite images alone.

Feasibility and execution plan

The first phase is designed to establish whether the project is measurable and whether a credible outcome study is possible. A larger analysis will proceed only when the registry, imagery pilot, and data agreements support it.

Period	Work and decision
Months 1 and 2	Audit operational sources; reconstruct pilot coverage and active dates; test image detectability. Decide whether imagery adds usable evidence beyond documentary records.
Months 3 and 4	Construct baseline GIS access measures; validate annotations; seek operator or healthcare data agreements and required ethics approval. Define outcomes and comparison units before estimation.
Months 5 and 6	Run the feasible analysis, robustness checks, and heterogeneity tests. Produce a documented registry, measurement protocol, research manuscript, and access maps within license limits.

Fallbacks and decision rules

If relevant infrastructure is not reliably visible, imagery will serve only as environmental context and the registry will remain document-based. If precise service boundaries cannot be verified, uncertain locations will be flagged or excluded rather than assigned broad city or county exposure. If patient or operator data cannot be obtained, the deliverable will be an access-measurement and feasibility study with no claims about receipt, adherence, or health.

Data governance and noncommercial use

This is noncommercial academic research at KAIST, separate from TalentSeeker's business activities. Planet access and U.S. outcome-data access have not yet been secured. Licensed imagery will be used and shared only under applicable terms; public outputs will contain permitted derived material and source documentation. Patient-level work will require appropriate institutional review, data-use agreements, secure processing, and aggregate reporting. No patient records or commercial datasets will be published with the research materials.

Researcher preparation

I am a Ph.D. student in Management Information Systems in KAIST's Business and Technology Management program, advised by Professor Chul Ho Lee. My preparation combines OSM-supported road detection and physics/XGBoost speed-recommendation work, behavioral research on 444 personal-mobility users and non-users using PLS-SEM and multi-group analysis, and a Yuseong District-supported study linking Tashu bike-sharing, transit, spending, and population-movement data.

My healthcare work includes development and study of an LLM/RAG chatbot with Jeonbuk National University Hospital and the ICIS 2026 proceedings paper, Thought-Bubbles in Therapy, with Donggyu Min, Antino Kim, and Chul Ho Lee [14]. I also serve as Chief Data Officer at TalentSeeker and teach data science at Baekseok University. These experiences support the project's technical and behavioral components; U.S. operator and patient data access remains to be established.

References

Selected scholarly literature and primary operational and product sources. Web sources checked 30 October 2026.

[1] Haidari, L. A., et al. (2016). The economic and operational value of using drones to transport vaccines. Vaccine, 34(34), 4062–4067. doi: 10.1016/j.vaccine.2016.06.022

[2] Nisingizwe, M. P., et al. (2022). Effect of unmanned aerial vehicle (drone) delivery on blood product delivery time and wastage in Rwanda: a retrospective, cross-sectional study and time series analysis. The Lancet Global Health, 10(4), e564–e569. doi: 10.1016/S2214-109X(22)00048-1

[3] Manne, N., Coustasse, A., Kimble, C., Simonton, D., & Pile, T. (2026). Drone-supported medication delivery as a rural pharmacy access strategy: A scoping review of logistics, regulation, and pharmacist integration. Journal of the American Pharmacists Association, advance online publication, 103553. doi: 10.1016/j.japh.2026.103553

[4] Levesque, J.-F., Harris, M. F., & Russell, G. (2013). Patient-centred access to health care: conceptualising access at the interface of health systems and populations. International Journal for Equity in Health, 12, 18. doi: 10.1186/1475-9276-12-18

[5] Strong, D. M., et al. (2014). A Theory of Organization-EHR Affordance Actualization. Journal of the Association for Information Systems, 15(2), 53–85. doi: 10.17705/1jais.00353

[6] Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. Journal of Econometrics, 225(2), 200–230. doi: 10.1016/j.jeconom.2020.12.001

[7] Amazon Pharmacy (2024, July 25). Prescription drone-delivery pilot report submitted to the Texas State Board of Pharmacy, August 2024 board materials, item D.1. Source

[8] Amazon. Amazon Pharmacy prescription drone-delivery announcement and subsequent College Station closure notice. Source

[9] Intermountain Healthcare (2022, October 4). Announcement of Zipline and Intermountain drone deliveries beginning in the Salt Lake Valley. Source

[10] Planet. PlanetScope imagery documentation. Source

[11] Planet. SkySat imagery documentation. Source

[12] Planet. Education and Research Program. Source

[13] Planet Support. SkySat data and Tasking Dashboard access in the Education and Research Basic Program. Source

[14] Lee, W., Min, D., Kim, A., & Lee, C. H. (2026). Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts. ICIS 2026 Proceedings, Healthcare and Wellbeing, 19. Source